

Chapter 3 – Stoichiometry

3.1. The Mole and Molar Mass

1. Know the definition of **Avogadro's number** and its relationship to the definition of a **mole**.
2. Know the definition of **molar mass** and how to calculate **formula weight** for a compound given the type and number of atoms in the compound formula.

3.2. Stoichiometry and Compound Formulas

1. Be able to determine **percent composition** from a **compound formula**.
2. Be able to define and distinguish **molecular formula** and **empirical formula** and determine an empirical formula from a molecular formula or a percent composition.
3. Be able to define **water of hydration** and identify a **hydrated compound**.

3.3. Stoichiometry and Chemical Reactions

1. Be able to define **reactant** and **product** and identify them in a specified **chemical reaction**.
2. Be able to represent a chemical reaction by a **balanced chemical equation** including **physical state labels** for the reactants and products.
3. Be able to determine **stoichiometric factors** (ratio of **balancing coefficients**) from a balanced chemical reaction and use these factors in a **mole-mole calculation** when one reactant is completely consumed.
4. Be able to use the formula weight for reactants and products and stoichiometric factors from a balanced chemical equation to perform a **mass-mole** or a **mass-mass calculation** when one reactant is completely consumed.

3.4. Stoichiometry and Limiting Reactants

1. Be able to define **limiting reactant** and **excess reactant** and identify both for a specified quantity of reactants.
2. Be able to define **theoretical yield**, **actual yield**, and **percent yield** and determine theoretical yield for a specified quantity of reactants.
3. Be able to determine amount of excess reactant remaining for a specified quantity of reactants.

3.5. Chemical Analysis

1. Be able to use **combustion analysis** to determine the empirical formula for a compound containing only carbon, hydrogen, and oxygen.

Chapter 4 – Solubility and Solutions

4.1. Types of Chemical Reactions

1. Be able to define and identify **combustion reactions**, **decomposition reactions**, **single displacement reactions**, and **double displacement reactions**.

4.2. Aqueous Solutions

1. Understand a **solution** as a homogeneous mixture and the role of **solvent** and **solute** as components of a solution.
2. Know that **aqueous** refers to a solution for which water is the solvent.
3. Understand the solute-solvent interactions implied by referring to the solute as **solvated** and recognized that **hydrated** is the special case of solvated where the solvent is water.
4. Understand an **electrolyte** as a substance that dissociates into ions in aqueous solution and recognize the difference between **strong electrolytes** and **weak electrolytes**.
5. Distinguish between **soluble** and **insoluble** as a physical property for a substance and recognize a **precipitate** as an insoluble product of soluble reactants.
6. Know and be able to apply the **solubility rules** in the Chapter 4 course notes to predict solubility, whether the mixing of two solutions will produce a precipitate, and if so, what the precipitate is.

4.3. Reactions in Aqueous solutions

1. Be able to define and distinguish **saturated solution**, **unsaturated solution**, and **supersaturated solution**.
2. Be able to define and recognize a **precipitation reaction**.
3. Understand the relationships between a **molecular reaction**, **complete ionic reaction**, and **net ionic reaction** representations of a precipitation reaction, and be able to use solubility rules to write all three representations for a specified precipitation reaction.
4. Be able to identify **spectator ions** in a complete ionic equation.
5. Be able to define and identify **acids**, **bases**, and **salts**.
6. Be able to distinguish **strong acids** and **weak acids** and know the common strong acids.
7. Be able to distinguish **strong bases** and **weak bases** and know the common strong bases.
8. Be able to define and identify **monoprotic acids**, **diprotic acid**, and **triprotic acids**.
9. Be able to recognize and determine the molecular equation, complete ionic equation, and net ionic equation for an **acid-base neutralization reaction**.
10. Be able to define a **gas-forming reaction** and identify the gas form when a **carbonate**, **hydrogen carbonate**, **sulfite** or **hydrogen sulfite** salt reacts with an acid.

4.4. Oxidation-Reduction Reactions

1. Know that **oxidation** is the loss of one or more electrons, **reduction** is the gain of one or more electrons, and recognize that oxidation cannot take place without reduction.
2. Know that the **oxidizing agent** is the reactant that causes oxidation and recognize that it does so by gaining electrons and being reduced.

3. Know that the **reducing agent** is the reactant that causes reduction and recognize that it does so by losing electrons and being oxidized.
4. Understand that a **half-reaction** is a theoretical construct showing an oxidation-only reaction where electrons appear as a product or a reduction-only reaction where electrons appear as a reactant.
5. Recognize that a whole oxidation-reduction reaction is obtained by adding an oxidative half-reaction to a reductive half-reaction in a way that the reactant and product electron cancel in the net reaction.
6. Understand that **oxidation numbers** are a theoretical construct used to keep track of electrons transferred in an oxidation-reduction reaction and know how to assign oxidation numbers using the scheme given in the chapter 20 course notes.
7. Recognize that an **oxidation number change** for an element as reactants are transformed into products is the characteristic that identifies a reaction as oxidation-reduction.

4.5. Solution Concentration and Molarity

1. Understand **concentration** as a quantitative measure of the amount of a substance dissolved in a specified amount of solution or solvent.
2. Recognize **molarity**, defined by $\frac{\text{moles of solute}}{1 \text{ L of solution}}$, as a concentration unit, denoted by M , and represented by a square bracket notation, [solute].
3. Be able to use the molarity of a strong electrolyte solution and the formula of the ionic compound to determine the molarity of the individual ions arises from the dissolution of the ionic compound.
4. Be able to use molarity as a conversion factor allowing moles of solute to be determined given a specifies volume of solution or volume of solution to be determined given a specified amount of solute.
5. Be able to determine solution concentration for solutions prepared by the **direct addition** or **dilution** method using **volumetric glassware**.
6. Be able to use solution volume and concentration for reactants and/or products and stoichiometric factors from a balanced chemical equation to perform a **mass-mole** or a **mass-mass calculation**.
7. Be able to define **indicator**, **equivalence point**, **standardization**, and **primary standard** and explain their significance in **acid-base titration**.
8. Be able to determine acid or base concentration from titration data.
9. Be able to determine acid molecular weight from titration data for a specific mass of the acid.